organisms in the specimen without altering their concentration. It mainly contains buffers and salts. Example: Stuart's transport medium that lacks Carbon, Nitrogen and organic growth factors. Other examples of transport media are Cary Blair and Amies.

Infobits

Viral Transport Medium is used to carry a specimen containing viruses. Universal Transport Viral Medium (UTVM). This liquid medium is stable at room temperature. It is used for collection, transport, and maintenance and long term freeze storage of viruses.

Exceptions in cultivation of microbes in artifical medium

Some bacteria like *Mycobacterium leprae* and *Treponema pallidum* cannot be cultivated in artificial medium.

ix) Media used for isolation of fungi

Apart from the bacteriological media, fungal media are used to study fungal morphology pigmentation and sporulation. Sabouraud's Dextrose Agar (SDA) is used as a common medium to isolate fungus. There are several other important fungal media used for fungal cultivation. Examples Niger Seed Agar and Potato Dextrose Agar

HOTS

1. Which medium is used to carry the sample when the sick person is unable to come to the laboratory?
2. Give a special medium to check the growth of anerobes in a burn wound infection with dead tissues.

5.3 Pure Culture

In nature, microorganisms usually exist as complex multispecies community. A single species has to be characterized in order to know the morphology, pathogenicity and molecular genomic pattern of the organism. For characterizing a species we have to isolate the organisms in pure form. Pure culture or axenic culture is a culture containing only one type of organism. The descendents of a single organism in pure culture is called a strain. A strain forms a single colony. Colony is a cluster of microorganisms in which all the characters of the family remain same. With the advent of the pure culture techniques many microorganisms are being identified.

5.3.1 Methods Employed in the Isolation of Microorganisms

Though there are many methods designed for isolation of microorganisms, pour plate method, spread plate method and streak plate method are widely used in the field of Microbiology.

i) Pour plate method

- It is the used for the isolation and counting of colony forming bacteria in the specified sample.
- In this technique a sample is diluted several times to reduce the density of the microbial population.
- A very small amount of diluted sample (1ml or 0.1ml) is mixed with the molten agar at a temperature of 45°C.
- The mixture is poured into the sterile petridish (In 1887, Juluis Richard Petri, a worker in Koch's laboratory, designed the Petriplate.) in an aseptic condition

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